Cyclistic Bike-Share Analysis - Yazan Muhieddin

Understanding the Differences Between Casual Riders and Annual Members

Analysis Period: January–May 2025
Tools: Google BigQuery (SQL), Google Sheets, Tableau Public
Dataset: Divvy Bike-Share Trip Data
Total Cleaned Rides: 1,430,076

Interactive Dashboard: View on Tableau Public

Executive Summary

This case study analyzes bike-share usage patterns to understand how casual riders and annual members use Cyclistic bikes differently. The purpose of the analysis is to identify opportunities for converting casual riders into annual members through data-driven marketing strategies.

Using publicly available Divvy bike-share trip data from January through May 2025, I analyzed 1,430,076 cleaned rides using Google BigQuery, Google Sheets, and Tableau Public.

The analysis revealed three important findings:

1. Casual riders take longer trips. Casual riders averaged 22.25 minutes per ride, compared with 11.44 minutes for annual members.
2. Annual members demonstrate stronger weekday usage. Member rides were concentrated around typical commuting periods, while casual riders showed relatively greater weekend activity.
3. Ridership increased substantially toward May. Both rider groups recorded more trips in the later months of the analysis period, particularly during April and May.

These findings suggest opportunities for Cyclistic to promote annual memberships through weekend-focused marketing, campaigns timed around increasing spring ridership, and messaging tailored to different riding behaviors.

1. **Ask — Define the Business Task**

Business Background

Cyclistic is a fictional bike-share company based in Chicago. It operates a fleet of bicycles available to customers through single-ride passes, full-day passes, and annual memberships.

The company distinguishes between two customer categories:

- Casual riders: Customers who purchase single-ride or full-day passes.

- Annual members: Customers who purchase an annual membership.

Cyclistic's marketing team believes that increasing the number of annual members is an important opportunity for long-term business growth.

Rather than focusing exclusively on attracting new customers, the company wants to explore how existing casual riders could be encouraged to become annual members.

Business Question

How do annual members and casual riders use Cyclistic bikes differently?

Business Objective

The objective of this analysis is to compare riding patterns between casual riders and annual members, identify meaningful differences in their behavior, and develop three recommendations that Cyclistic's marketing team could use to encourage membership conversions.

Key Stakeholders

Lily Moreno — Director of Marketing: Responsible for developing marketing strategies to increase annual memberships.

Cyclistic Marketing Analytics Team: Responsible for analyzing customer behavior and supporting data-driven recommendations.

Cyclistic Executive Team: Responsible for evaluating and approving proposed marketing strategies.

2. **Prepare — Collect and Understand the Data**

Data Source

The analysis uses publicly available historical bike-share trip data provided by Divvy, Chicago's bike-share system.

Original Data Source: Divvy Trip Data

The dataset serves as a real-world proxy for the fictional Cyclistic company described in the Google Data Analytics case study.

Analysis Period

The dataset covers five months:

January 1, 2025 – May 31, 2025

I downloaded the monthly trip-history files for January, February, March, April, and May 2025 and imported the data into Google BigQuery for cleaning and analysis.

The scope was limited to these five months because of file upload and processing constraints encountered during the project.

Dataset Information

The trip-history data includes fields such as:

Field	Description
ride_id	Unique identifier for each recorded ride
rideable_type	Type of bicycle used
started_at	Date and time the ride began
ended_at	Date and time the ride ended
start_station_name	Starting station
end_station_name	Ending station
start_lat / start_lng	Starting coordinates
end_lat / end_lng	Ending coordinates
member_casual	Rider category: member or casual

These fields make it possible to analyze trip duration, riding frequency, and usage patterns across different dates and times.

Data Reliability and Limitations

Divvy provides real trip-history records, making the dataset appropriate for exploring observed riding patterns.

However, there are important limitations:

- The data covers only January–May 2025, not a complete calendar year.
- Trip records represent individual rides rather than unique customers.
- The dataset does not provide customer identities, membership conversion histories, or individual purchasing decisions.
- Trip data alone cannot establish why customers ride or whether a particular marketing campaign would be profitable.
- Divvy is a real bike-share service, while Cyclistic is a fictional company used for the case study.

These limitations are considered when interpreting the findings and developing recommendations.

3. **Process — Clean and Prepare the Data**

Tools Used

Google BigQuery (SQL) was used to combine, validate, clean, and analyze the monthly trip data.

Google Sheets was used to organize aggregated results and create exploratory charts.

Tableau Public was used to develop the final interactive dashboard.

Data Cleaning Process

After importing the monthly files into BigQuery, I used SQL to inspect the records and prepare the dataset for analysis.

The cleaning process included the following steps:

Step 1: Combine monthly datasets

Combined the January–May 2025 trip records into one dataset to allow consistent analysis across the five-month period.

Step 2: Check data quality

Examined the data for duplicate ride identifiers, missing essential fields, and problematic ride durations.

The validation checks found no duplicate ride IDs and no missing values in the essential ride ID, timestamp, or rider-type fields.

Step 3: Calculate ride duration

Calculated each trip's duration using the difference between its ending and starting timestamps, expressed in minutes.

Step 4: Filter very short rides

Excluded 32,392 rides lasting less than one minute. This was an analysis-quality rule intended to remove trips too short to be meaningful for the ride-duration comparison. It does not necessarily mean that every excluded trip was erroneous.

Step 5: Confirm the analysis period

Applied the January 1–May 31, 2025 date boundary, excluding 53 additional records that fell outside the intended local-date period.

Step 6: Prepare analysis variables

Used the cleaned timestamps to derive day-of-week, month, and hour-of-day information for comparison between rider types.

Cleaning Results

Metric	Ride Count
Initial records	1,462,521
Rides under one minute excluded	32,392
Additional records outside analysis dates	53
Final cleaned dataset	1,430,076

The final cleaned dataset was used consistently for the rider comparison, weekday, monthly, and hourly analyses.

4. Analyze — Identify Trends and Patterns

The analysis focused on four areas: overall rider activity, ride duration, weekday usage, and monthly and hourly riding patterns.

Finding 1: Annual Members Account for Most Recorded Rides

Rider Type	Total Rides	Share of Rides
Annual members	1,014,978	71.0%
Casual riders	415,098	29.0%
Total	1,430,076	100%

Annual members accounted for approximately 71% of all recorded rides during the analysis period.

This indicates that members generated a larger share of trip activity than casual riders.

However, this does not establish the number of unique customers in either group or how often individual customers rode.

Finding 2: Casual Riders Take Longer Trips

Rider Type	Average Duration	Median Duration
Casual riders	22.25 minutes	10.75 minutes

Annual members	11.44 minutes	7.92 minutes

Casual rides lasted approximately 1.94 times as long on average as member rides.

This suggests that casual riders may be more likely to use bicycles for longer recreational outings, while members may be more likely to take shorter, routine trips.

The difference between average and median duration also suggests that longer rides influence the average, especially among casual riders.

Trip duration alone cannot confirm each rider's purpose, but the pattern is useful for developing marketing hypotheses.

Finding 3: Members Have Stronger Weekday Usage

The day-of-week analysis showed that annual members recorded more rides than casual riders on every day of the week.

Member activity was particularly strong on weekdays, with Thursday recording the highest member total at 170,136 rides.

Casual ridership reached its highest daily total on Saturday, with 85,871 rides.

Day	Casual Rides	Member Rides
Monday	50,147	145,742
Tuesday	48,616	162,395
Wednesday	44,366	152,464
Thursday	54,409	170,136
Friday	74,259	158,541
Saturday	85,871	129,259
Sunday	57,430	96,441

Casual riders showed a relatively stronger preference for weekends, while members demonstrated more consistent weekday activity.

This pattern is consistent with the possibility that many members use bike-share services for transportation and commuting, whereas casual riders may be more interested in recreational or occasional use.

Finding 4: Ridership Increased Toward May

Month	Casual Rides	Member Rides
January	23,453	112,352
February	27,051	122,118
March	83,064	208,500
April	105,528	257,962
May	176,002	314,046

Both groups experienced substantial increases in riding activity between January and May.

Casual ridership increased from 23,453 trips in January to 176,002 in May, while member ridership rose from 112,352 to 314,046.

The increases may be related to warmer weather and greater interest in outdoor activities, but the dataset does not directly measure weather or customer motivations.

Because only five months were analyzed, the results should be described as a January–May trend rather than proof of a complete annual seasonal pattern.

Finding 5: Members Show Stronger Commuting-Hour Patterns

Hourly analysis showed that member ridership was especially strong during typical morning and late-afternoon commuting periods.

Member rides reached:

- 75,865 rides at 8 AM
- 99,611 rides at 4 PM
- 111,058 rides at 5 PM

Casual riding activity was more concentrated in the afternoon and early evening, with the highest hourly total of 41,837 rides at 5 PM.

The presence of both morning and evening peaks among members supports the hypothesis that routine transportation is an important part of annual-member usage.

Casual riders' comparatively longer trips and weekend activity suggest a different mix of potential riding purposes.

However, these observations do not directly identify the purpose of any individual ride.

5. Share — Communicate the Findings

Interactive Tableau Dashboard

To communicate the results, I developed an interactive dashboard using Tableau Public.

Dashboard: Cyclistic Bike-Share Analysis | January–May 2025

The dashboard includes four visualizations:

1. Average Ride Duration by Rider Type

A bar chart comparing the average ride duration of casual riders and annual members.

2. Weekday Riding Patterns

A grouped bar chart comparing the number of rides by rider type for each day of the week.

3. Monthly Ridership Trends

A line chart illustrating how casual and member ride counts changed from January through May 2025.

4. Hourly Riding Patterns

A line chart comparing the distribution of rides throughout the day for both rider categories.

Summary of Key Insights

The visualizations highlight three main differences between the rider groups:

- Casual riders generally take longer trips and show relatively greater weekend activity.
- Annual members account for most rides and demonstrate stronger weekday and commuting-hour patterns.
- Both groups recorded substantial ridership growth from January through May, creating opportunities to test marketing campaigns during periods of increasing demand.

These findings provide a foundation for developing targeted marketing strategies.

6. Act — Business Recommendations

Based on the analysis, I developed three recommendations for Cyclistic's marketing team.

Recommendation 1: Promote Annual Memberships to Weekend Riders

Supporting evidence: Casual ridership was highest on Saturdays, and casual riders took substantially longer trips on average than annual members.

Recommended action: Test marketing campaigns that explain the benefits of annual membership to customers who use bike-share services for weekend outings and recreational activities.

Potential tactics include weekend-oriented membership messaging, comparisons of pass costs with membership costs, and limited-time membership promotions.

Expected opportunity: Encourage casual riders who use bicycles repeatedly for leisure activities to consider whether an annual membership offers better value.

How to measure success: Compare membership conversion rates and acquisition costs between campaign participants and a suitable control group.

Recommendation 2: Launch Campaigns Ahead of Increasing Spring Ridership

Supporting evidence: Casual rides increased from 23,453 in January to 176,002 in May.

Recommended action: Test a spring marketing campaign beginning ahead of the observed March–May increase in bike-share activity.

Marketing could highlight membership benefits as customers begin riding more frequently during warmer months.

Expected opportunity: Reach potential members during a period when casual bike-share activity is increasing.

How to measure success: Track membership sign-ups, campaign conversion rates, and acquisition costs, ideally using an A/B test to separate campaign effects from underlying seasonal changes.

Recommendation 3: Tailor Membership Messaging to Riding Behavior

Supporting evidence: Annual members demonstrated stronger weekday and commuting-hour activity, while casual riders showed relatively greater weekend usage and longer average trips.

Recommended action: Develop different marketing messages for different riding needs.

For customers interested in routine transportation, emphasize the convenience of an annual membership. For recreational riders, emphasize potential membership value for repeated leisure trips.

Cyclistic could also test promotions during high-activity casual-riding periods, particularly Friday through Sunday and afternoon hours.

Expected opportunity: Make membership marketing more relevant to different types of riders instead of relying on one general message.

How to measure success: Evaluate campaign conversion rates, membership retention, and return on marketing investment.

Because the public dataset does not identify individual customers, person-level targeting would require separate, appropriately permissioned customer data.

Conclusion

The purpose of this analysis was to understand how casual riders and annual members use Cyclistic bikes differently and identify potential opportunities to increase annual memberships.

After analyzing 1,430,076 cleaned rides from January through May 2025, I found that annual members generated the most recorded rides and showed stronger weekday and commuting-hour activity. Casual riders took longer trips and showed relatively greater weekend usage.

Ridership also increased substantially toward May, suggesting an opportunity to test membership campaigns during periods of growing bike-share activity.

Based on these findings, I recommend that Cyclistic:

1. Test annual-membership promotions aimed at weekend and recreational riders.
2. Launch targeted campaigns ahead of the observed spring ridership increase.
3. Develop membership messaging tailored to different riding patterns.

These recommendations are hypotheses informed by observed trip behavior, not proven solutions. Controlled campaign testing, membership pricing information, and customer conversion data would be needed to evaluate their effectiveness.

Overall, this project demonstrates how SQL-based data preparation, exploratory analysis, and interactive visualization can support business decision-making.

Project Tools and Skills Demonstrated

Google BigQuery (SQL)

- Data import and consolidation
- Data validation and cleaning
- Date and time analysis
- Aggregation and grouping
- Rider behavior comparisons

Google Sheets

- Data organization
- Pivot tables
- Exploratory charts

Tableau Public

- Data visualization
- Dashboard design
- Presentation of business insights

Business Analytics

- Defining a business question
- Interpreting customer behavior
- Identifying limitations
- Developing actionable recommendations
- Communicating findings to stakeholders

Data Sources and References

Divvy Bike-Share System. Publicly available trip-history data, January–May 2025. https://divvy-tripdata.s3.amazonaws.com/index.html

Google Data Analytics Professional Certificate. Cyclistic Bike-Share Case Study.

Tableau Public Dashboard. Cyclistic Bike-Share Analysis | January–May 2025. https://public.tableau.com/views/CyclisticBike-ShareAnalysisJanuaryMay2025/Dashboard1

Note: Cyclistic is a fictional company used for educational purposes. This independent portfolio analysis uses publicly available Divvy trip-history data and is not an official analysis or recommendation issued by Divvy.